
Paper III: The Alignment Scaling Problem

FULL TITLE	The Alignment Scaling Problem: Why External AI Safety Approaches Cannot Scale With Recursive Capability
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This paper demonstrates that current AI alignment approaches produce alignment scaling exponents of approximately zero, meaning safety degrades relative to capability as recursive depth increases. If AI capability scales super-linearly through recursive self-correction (confirmed in 95.6% of tested configurations), but alignment constraints such as RLHF, constitutional rules, and output filters operate externally to the reasoning process, then a growing capability-alignment gap is mathematically inevitable. The experiments use a blinded evaluation protocol across multiple frontier models to measure alignment quality as a function of inference-time reasoning depth.

Experiments

All experiment scripts and results are in [experiments/](#).

Note: Papers IVa through IVd share this same experiment suite (alignment-scaling v1 through v5). See also Paper-IV-a, IV-b, IV-c, and IV-d folders.

Links

- **OSF DOI:** <https://doi.org/10.17605/OSF.IO/6C5XB>
- **GitHub:** <https://github.com/MichaelDariusEastwood/arc-principle-validation>

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